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Abstract

Group music improvisation is universal across cultures, offering a window into real-time creative processes. Alpha oscillations in electroencephalography (EEG) during social interaction reflect individual role and between-partner task similarity. Alpha may also increase with more creative tasks, with mixed results for music improvisation. We examine how alpha power varies in improvisation compared to performing scored music, and how it interacts with the individual's musical role (starter or joiner in a turn-taking paradigm) and partner's task (score or improvisation). Two pianists alternated playing and listening to their partner, either following a score or improvising during two-EEG recording. Alpha power during listening increased then decreased, which was suppressed for improvisation compared to the score. During playing, alpha power was flat followed by an increase, which was exaggerated when improvising. These effects were enhanced for the joiner, especially when both players had the same task (score or improvisation). Furthermore, during joint improvisation, a player's role-related flexibility (alpha power difference between the two roles) was related to improvisation experience during listening and empathy level during playing. We propose that alpha coordinates creative freedom, attentional resources, and empathic affordance under social scenarios, highlighting the importance of role and task on interactive, goal-oriented co-creation enterprises.

Humans improvise daily during conversation, cooking, dancing, or playing sports, and improvising musically is a universal human activity present across cultures. Music improvisation is also a highly trainable skill, and expert musical improvisers show marked differences from novices in their brain activity (Limb and Braun 2008; Beaty 2015; Lopata et al. 2017). In many performances, the musical materials such as notes, timing, and dynamics are preconceived beforehand in a written score. During improvisation, however, these materials are partly or entirely free from predetermined constraints, requiring the brain to devote additional resources to making creative decisions on the fly while monitoring the outcome.

Playing music with others requires a complex network of brain systems enabling coordination of precise muscular movements in time, processing of auditory and tactile feedback, and adjustment of future plans. The process of listening, acting, and adapting in the moment follows the perception-action cycle, where the frontal lobe of the brain plays a substantial role (Blakemore and Decety 2001; Ridderinkhof et al. 2004; Hari and Kujala 2009). The perception-action cycle is an important framework to understand 'joint action' that occurs between individuals to achieve a common goal (Sebanz et al. 2006; Novembre et al. 2014; Sebanz and Knoblich 2021). In group music improvisation, people have to temporally coordinate joint actions while balancing their internal attention to idea generation and their external attention to listening and integrating feedback, enabling continuous adaptation to an evolving musical context (Keller et al. 2014). Thus, group music improvisation provides an ecologically valid lens for investigating real-time creative processes and joint action in the brain (D'Ausilio et al. 2015; Czeszumski et al. 2020; Setzler and Goldstone 2020; Setzler and Goldstone 2022). Our interest is in (1) identifying neural correlates of a creative mindset during music improvisation that is distinct from performing scored music, and (2) understanding how joint musical goals and individual roles differently influence creative performance with a partner during listening and playing.

Alpha oscillations (8-13 Hz) are one neural correlate of sensorimotor and cognitive processing, typically observed non-invasively in electroencephalography (EEG) or magnetoencephalography (MEG). The decrease of alpha power with movement or sensory input is referred to as the event-related desynchronization (ERD), and is followed by alpha power increase, or event-related synchronization (ERS) after the movement or sensory input terminates (Pfurtscheller 1992; Pfurtscheller et al. 1996; Babiloni et al. 2002). Typically, alpha ERD begins a few seconds before an isolated voluntary movement or an external stimulus onset, reaching a minimum just after onset (Neuper and Pfurtscheller 1996; Pfurtscheller and Lopes da Silva 1999; Klimesch 2012), and rebounds within 1-2 seconds after movement termination (Hari and Salmelin 1997; Pfurtscheller and Neuper 2003). In addition, observing the actions of others or imagined actions results in alpha ERD/ERS, although the magnitude is usually smaller than self-performed actions (Hari and Salmelin 1997; Pfurtscheller and Neuper 1997; Pfurtscheller et al. 1999; McFarland et al. 2000; Neuper and Pfurtscheller 2001; Babiloni et al. 2002). Furthermore, prior motor training enhances the alpha ERD during observation (Cannon et al. 2014), and listening to familiar music induces alpha ERD in frontal and posterior areas compared to listening to unfamiliar music (Malekmohammadi et al. 2023).

Regarding cognitive processes, mounting evidence has shown that alpha oscillations increase as attention shifts from external (sensory processing) to internal (mental imagery). This has led to recent interpretations of the functional significance of alpha ERS as top-down inhibition of brain areas that are not actively involved in processing (Klimesch et al. 2007). Meanwhile, brain regions exhibiting alpha ERD reflect areas of active processing (Neuper and Pfurtscheller 2001; Cooper et al. 2003; Cooper et al. 2006; Klimesch et al. 2007; Mazaheri et al. 2014). Together, these findings indicate modulation of alpha oscillations by active, observed, and imagined movement in conjunction with familiarity and attention.

Alpha oscillations are also related to creative cognition and have been investigated through the lens of divergent thinking or alternate use tasks (Grabner et al. 2007; Fink et al. 2009b; Schwab et al. 2014; for review see Fink and Benedek 2014). Alpha ERS correlates positively with the degree of creativity involved in execution of a task, increases with prior training in a creative task, and is larger for

individuals with higher levels of creativity (Cooper et al. 2003; Fink et al. 2009a; Fink and Benedek 2014). Evidence indicates that frontal alpha ERS reflects internal processing related to creative idea generation, parietal alpha ERS reflects attention oriented toward long-term memory search and retrieval, and occipital alpha ERS reflects suppression of visual distractions (Fink and Benedek 2014). However, critics have pointed out that these tasks, for example requiring participants to "come up with all possible uses for a brick," have little connection to real-world creative processes (Dietrich 2007). In contrast, music improvisation is practiced widely in various cultural traditions, and therefore is an ecologically natural activity of human creativity.

To date, studies involving solo music improvisation have indeed demonstrated the importance of alpha activities, but results have been contradictory (Adhikari et al. 2016; Lopata et al. 2017; Boasen et al. 2018). Some studies found increased alpha power during improvisation compared to the control condition (Lopata et al. 2017; Boasen et al. 2018), but other studies found the opposite (Adhikari et al. 2016). These divergent results may be due to different definitions of the improvisation or control task, which is also prevalent in any neuroscientific studies on improvisation. For example, some studies require performers to memorize their improvisations to be replayed later, because the control condition was defined as imitation (Limb and Braun 2008; for review see Beaty 2015). In addition, there are more factors that affect improvisational behavior and their neural correlates. Experienced improvisers played material of higher creative quality as rated by experts (Beaty et al. 2013). Attention-related brain areas differ between expert improvisers and non-musicians (Berkowitz and Ansari 2010). Alpha power differences between improvisation and control conditions were only significant for participants with music or improvisation experience (Lopata et al. 2017). This points to a possible link between improvisation training and alpha power modulation during improvisation. However, these studies have not yet addressed how joint goals and individual roles affect improvising with a partner.

Joint action studies have found that individual role, social context, and coordination affect alpha power modulation. For example, synchronized movement between two people results in greater alpha ERD in centro-parietal and sensorimotor areas than un-coordinated movement (Tognoli et al. 2007;

Naeem et al. 2012), which was further enhanced if musical duet partners were familiar with each other's parts (Novembre et al. 2016) and for the leader whose tapping timing was ahead of their partner during a tapping synchronization task (Konvalinka et al. 2014). There are a few EEG/MEG studies that have investigated music improvisation in a joint context. However, most have asked players to improvise simultaneously and looked at inter-brain synchrony or phase correlation between brain signals (Sänger et al. 2012; Müller et al. 2013; Müller and Lindenberger 2019) rather than the individual's alpha activity. Only one MEG study (Yoneta et al. 2022) asked two pianists to alternate listening and improvising in a turn-taking manner to investigate the individual's alpha related to leader/follower roles. When players served in the follower role, they had larger alpha ERS than when serving as a leader. This may be related to findings from joint coordination studies showing followers exhibit more adaptive behaviors than leaders (Fairhurst et al. 2014; Konvalinka et al. 2014; Timmers et al. 2014; Novembre et al. 2019; Washburn et al. 2021; Sabharwal et al. 2024), but none of these studies investigated improvisation. In an improvisation context, we may see behavioral adaptation differ depending on musical role, which may also be related to alpha power modulation.

Turn-taking successfully separates neural processes of encoding the partner's creative ideas during listening from playing one's own creative ideas. However, Yoneta et al. (2022) did not find an alpha power difference between improvisation and the control condition. Two main reasons may have contributed to this. First, their control condition required players to remember the partner's rhythm and imitate it while varying the notes, meaning that it involved some degree of improvisation. Second, players were asked to first imagine their improvised phrase before actually playing it in their improvisation condition. This may have likely introduced substantial working memory demands, which are reflected in the alpha responses, as mental imagery and working memory retention can also involve internally focused attention. Ideally, playing composed music from a score would be the best way to observe alpha differences from playing improvisation. Also, such differences may affect alpha activities during listening because encoding the partner's score playing would be easier than listening to the novel, improvised melody.

Lastly, the ability to understand others' actions is necessary for joint action. This ability, typically measured in individual trait empathy, has emotional and cognitive dimensions (Shamay-Tsoory 2011), that also affect alpha power modulation (Woodruff et al. 2011; Babiloni et al. 2012). For example, a study of saxophone quartets showed that larger alpha ERD in right frontal areas correlated with higher emotional empathy scores when musicians observed their own performance (Babiloni et al. 2012). Another study found that larger alpha ERD during action execution and observation was correlated with higher scores of perspective-taking trait empathy (Woodruff et al. 2011). In Novembre et al. (2014), when pianists were asked to synchronize with a recorded accompaniment part they practiced before, those with higher levels of perspective-taking score experienced increased motor excitability, demonstrated by the greater interference by transcranial magnetic stimulation (TMS). Thus, in joint improvised performance, the ability to model other's emotional and cognitive state, measured by empathy trait, may relate to neural resources available for more efficient and accurate predictions of their improvisation.

The present study aims to answer whether improvising with a partner influences alpha oscillations differently from playing a score, and how musicians balance generating creative ideas and paying attention to one's partner. We incorporated improvisation as part of a turn-taking duet paradigm very similar to our previous work which recorded simultaneous EEG from two piano players (Huberth et al. 2019). Our musical paradigm (Figure 1) was designed as a simple score consisting of multiple 6-note phrases. When players were assigned to improvise, they were instructed to vary the pitch content while maintaining the isochronous rhythm. This constraint of equating the actions between the improvisation and control (i.e. score) conditions is extremely important, because this avoids potential confounds of different movement timings.

Furthermore, turn-taking enables separating listening to others' parts from playing one's own part during joint music performance. In each trial, partners alternate playing the first four phrases (one player plays phrases 1 and 3, the other plays phrases 2 and 4), then both partners play the final two phrases in unison. Such alternation is typically called 'call-and-response,' a common way to create a natural musical interaction. Moreover, this results in a leadership role for the performer who starts and initiates the call

('Starter') and a follower role for the performer who responds afterward ('Joiner'). Our setup allowed us to test whether implicit leader/follower roles embedded in the musical structure as the Starter and Joiner would have any effect on improvisational behavior and alpha power. We did not give any instructions to perform differently in each role. Thus, our roles are different from more explicitly defined ones used throughout performance regardless of musical content (e.g. conductor and orchestra, jazz ensemble leader, first violin in a string quartet). We designed the experimental blocks to counterbalance roles, such that each participant performed as a Starter half of the time and a Joiner half of the time, varying by block.

Notably, our design also involved the combination of one performer playing the score and one improvising, in addition to both performers playing a score, or both improvising. Therefore, we had four different melodic conditions among the two partners. These combinations allowed us to look at whether task similarity between the duet partners had any effect on alpha power, which we found in our previous study for Starters performing with a computer versus human partner (Washburn et al. 2019). We suspected we may see a difference given previous studies showing alpha differences depending on familiarity (Novembre et al. 2016; Malekmohammadi et al. 2023). In our study, the score is familiar, whereas the pitches of the improvised phrases are unfamiliar. The hybrid scenario of one playing score and the other improvising, is common in musical settings around the world, including jazz, and often these roles will change frequently throughout a performance. In this case, since the partner's music is predictable, the improviser may be able to focus more easily on their creative ideas, in contrast to all performers improvising simultaneously. Thus, a player's creative and attentional demands may change depending on whether the partner is also improvising.

We had four hypotheses regarding the following questions: 1) Is there an alpha difference between the score and improvisation task while listening or playing? 2) How does the partner's task of score or improvisation affect alpha responses during listening and playing? 3) Does a player's role (Starter or Joiner) affect alpha responses or behavioral performance? (4) Do individual factors such as improvisation experience or empathy level affect alpha responses during joint improvisation?

Our first hypothesis is that during listening, participants assigned to the improvisation task will pay more attention to the partner, because they have to fit ideas into the ongoing musical context, leading to alpha power decrease. In contrast, participants assigned to the score task may pay less attention to the partner during listening, so alpha power may be larger than the improvisation task. During playing, we hypothesized that improvisation assignment may result in higher alpha power than the score (Lopata et al. 2017).

Our second hypothesis was that alpha power responses during listening and playing would likely further be affected by the partner's task of score or improvisation. During listening, if both partners are assigned to the improvisation task, we hypothesized that they would have a larger alpha decrease because they have to pay more attention to their partner's improvised melody than when their partner is playing the score. If a player is assigned to the improvisation task while their partner is assigned the score, we may see larger alpha power than when their partner is also improvising, since the partner's material is familiar and less relevant for one's own improvisation (Washburn et al. 2019). During playing, we suspected that it may be easier for the improviser to focus on their own creative ideas if their partner is assigned the score task, leading to larger alpha increase than if the partner is also improvising.

Third, we hypothesized that a partner's role (Starter or Joiner) may influence alpha power and/or behavioral performance. We anticipated that the Joiner may be more adaptive in their improvisation content based on previous findings (Fairhurst et al. 2014; Konvalinka et al. 2014; Timmers et al. 2014; Novembre et al. 2019; Washburn et al. 2021). We also predicted that alpha power may differ between the Starting and Joining roles given previous evidence (Konvalinka et al. 2014; Yoneta et al. 2022).

Fourth, we hypothesized that individual factors such as improvisation experience and empathy level may influence alpha power modulation during joint improvisation. Our primary interest was in how improvisation experience and empathy level may affect neural activity during joint improvisation, because prior studies did not look into improvisation. We expected more experienced improvisers will show advantages in creating higher quality improvisations based on behavioral evidence (Berkowitz and Ansari 2010; Beaty et al. 2013). We also predicted that alpha power modulation may be affected by

improvisation experience similar to a prior alpha study (Lopata et al. 2017). Regarding empathy level, we hypothesized that participants with higher empathy level may be more attentive to their partner while listening, based on the finding that individuals with higher levels of perspective-taking have larger alpha ERD during action observation of self and others (Woodruff et al. 2011; Babiloni et al. 2012).

Materials and Methods

Participants

Twenty-six pianists (thirteen pairs) participated in the study. They were recruited from the Stanford University community. These participants (13 female) ranged in age from 21 to 42 years (M = 26.3, SD = 4.5), and had 8 - 23 years of general musical experience (M = 16.2, SD = 4.2), with 14 subjects having 2 - 20 years of improvisation experience (M = 8.8, SD = 5.4). We did not collect further details about the nature of their improvisation experience. All participants were self-reportedly right-handed except one, who was ambidextrous. For the study, musicians were paired and completed all trials with the same partner. Some partners knew each other before completing the study. We included all 26 pianists in our final EEG analysis, but one pair's behavioral data (notes played and timing) was lost due to a technical problem, so behavioral analysis reflects 24 pianists. The Stanford University Institutional Review Board approved the study protocol and participants provided informed written consent. Pianists were paid $20/hour for their participation.

This sample size of 26 pianists is larger than that of our previous piano-duet alpha study (Washburn et al. 2019), which included 15 pianists. We used their results from the main effect of *Similarity* of melodic content for alpha power ($F(1,14) = 5.20$, $\eta_p^2 = 0.27$), which is the closest measure to our study, to calculate the required sample size. From the η_p^2 value we derived Cohen's f and for this factor we used a repeated-measures design (2 levels, 4 measurements). With the significance level of .05 and power of 0.8 (G*Power ver. 3.1.9.6) we obtained a minimum sample size of 6. This means that our sample size meets the minimum requirement.

Design and apparatus

Two M-Audio Axiom-61 MIDI keyboards were placed facing each other in a sound-isolated recording booth such that players could see each other while performing. A music stand with the score was placed in front of each player between the keyboards and low enough so that they could still see each other. The identical score containing both melodic parts was printed and given to both players. No explicit instructions were given to pianists to memorize the score or not.

Each trial required players to play through a short music consisting of six phrases, taking turns according to their role as the starting or joining player (Figure 1A). Trials began with 3 metronome clicks to indicate the tempo of each eighth note having 500 ms inter-onset-interval (IOI). For the first 4 phrases, players alternated each phrase, which consisted of 6 notes having roughly 500 ms IOI, with the allowable timing deviation for each key press only 125 ms (25%) longer or shorter. The starting player (called 'Starter') played the 1st and 3rd phrase, whereas their partner (called 'Joiner') played the 2nd and 4th phrase. Afterward, the Starter played the first 3 notes and was joined by the Joiner for the remaining 4 notes. Players used only their right hand for performance. Throughout each trial, a synthesized organ part played the bassline providing musical context, which was included to aid creativity during improvisation. The bassline was a typical harmonic progression in classical or jazz music (i.e., Tonic, Subdominant, Dominant, Tonic), and remained the same during all conditions (Score and Improv). Each note of the bassline started synchronized with the 1st note of its corresponding melodic phrase. After the final note onset of the trial, a silence of 1.5 - 2.5 seconds followed before the start of the next trial.

One block consisted of 16 test trials, preceded by 2 practice trials that were not analyzed. If errors were committed, the trial had to be repeated at the end of the block. Throughout each block, there were 2 possible *Task* conditions (Score, Improv) for each player; either they played the score as written or they improvised, resulting in four possible combinations for the pair (Figure 1B). In Improv conditions, pianists were required to keep the isochronous, monophonic eighth-note rhythm, but could use any pitches they wanted, resulting in 6 notes per phrase. For all conditions, the 5th and 6th phrases were played as written in the score. This also resulted in 2 *Similarity* conditions (Same, Diff), indicating

whether partners had the same *Task* or not. This means that if both partners played the score or both improvised, the block had a *Similarity* of Same. If one player improvised and the other played the score, the block had a *Similarity* of Diff. The order of *Task* over blocks was counterbalanced across the pairs. The pair iterated through the four blocks, called a super-block, without changing their *Role* as Starter or Joiner. After each super-block, partners switched roles and performed the corresponding *Task* conditions. This cycle was repeated 3 times, for a total of 6 super-blocks, 3 as Starter and 3 as Joiner for each player. Each block lasted roughly 6 minutes, resulting in about 2.5 hours of total recording time.

During each trial, the pitch of one note was altered for each player such that they heard a different pitch than the one played, in order to examine the feedback-related negativity (FRN), as described by Kim et al. (2025). This altered pitch could occur only on the 4th or 5th note of the phrase, and the played pitch was randomly increased or decreased by 2, 3, or 4 scale degrees in a C major diatonic scale (i.e. a white key note). Pianists were instructed to ignore the altered feedback and continue playing. The fifth and sixth phrase did not contain any alterations. The present study does not use any data from phrases containing altered feedback except for behavioral analysis of notes played during improvisation.

A Mac OS laptop running custom Max/MSP 7.0.1 software logged the note played, time, whether each trial ended with errors or no errors, and provided auditory feedback, which was played through two 3” Klein + Hummel M52 speakers, located on both sides of the pianists in the recording booth. The playback level was adjusted to roughly 80 dB SPL at the pianists’ head positions. The chosen piano timbre for all auditory feedback was the built-in AU DLS Synth sound. If a pianist assigned to the Score task played a wrong note, the trial was logged as an error trial, and that trial was added to the end of the block. No pitch was considered ‘wrong’ for improvising pianists, but notes could still be considered early or late. Trigger codes were sent from the Mac computer via USB to a custom Arduino Uno unit that translated the signals to those needed by the parallel port of an EEG acquisition computer, representing the *Task* condition (Score or Improv), metronome beats, notes played by pianists, pitches heard through auditory feedback, and trial outcome. Trigger timing was corrected to account for the latency between auditory stimulus and trigger recording, known to be 21 ms on this system (Huberth et al. 2019).

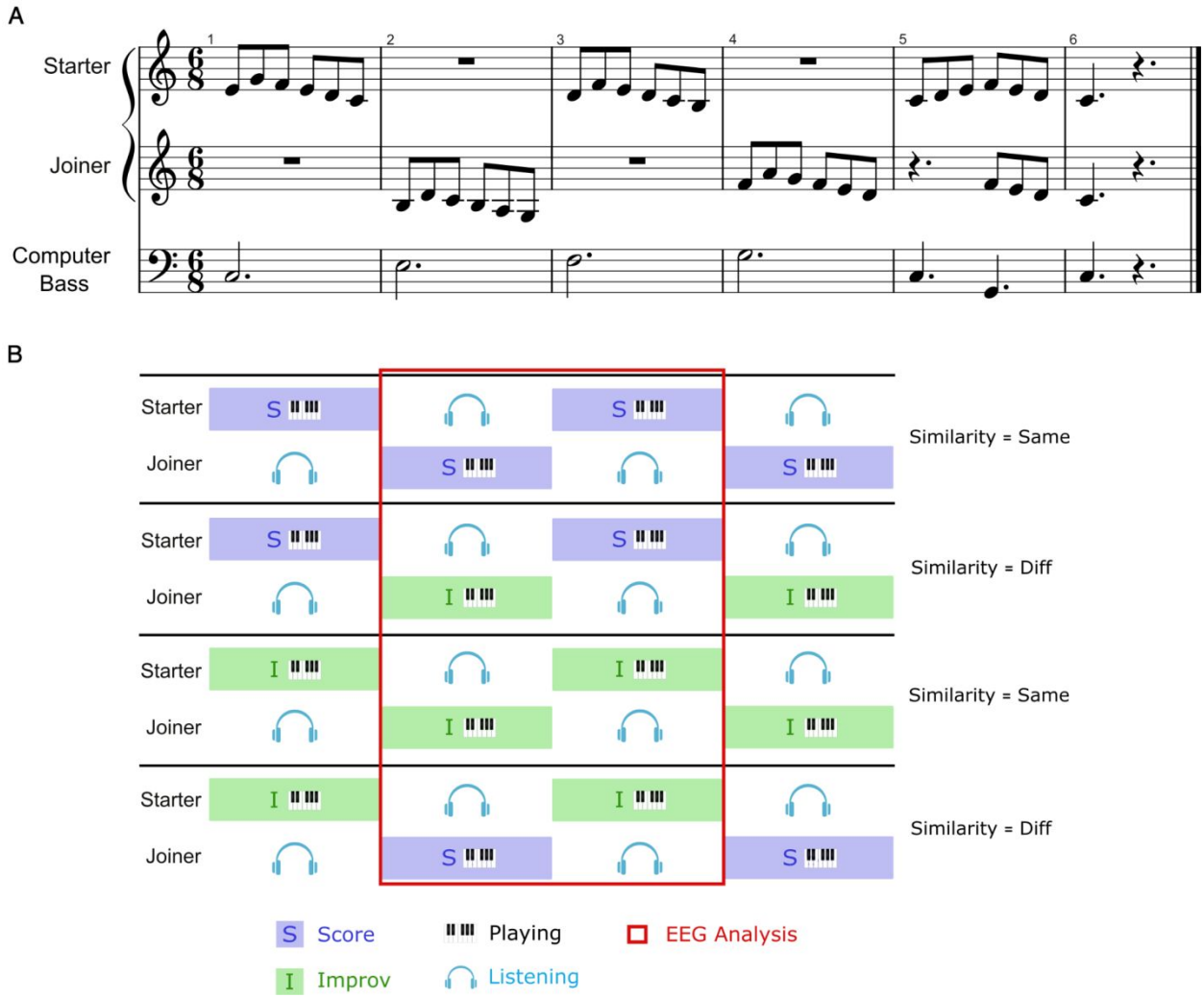

Figure 1. The duet study design and four conditions. A. The score when both players are assigned to play the score. Phrase number (1-6) indicated above the score. B. The four combinations of each *Role* (Starter, Joiner) and *Task* (Score (S), Improv(I)). During improvisation, pianists had to strictly adhere to the same rhythm and were instructed to freely choose one pitch per eighth note. After the 4th phrase, players followed the score as written for all conditions. If partners had the same *Task*, the trial had a *Similarity* condition of 'Same'; if one partner played the score and the other improvised, the trial's *Similarity* condition was 'Diff'. After playing each combination, partners switched Starter/Joiner roles and performed the respective *Task*. For EEG analysis, we focused on the 2nd and 3rd phrases, shown in the red box, because the previous and subsequent phrases exist for both roles, equating the movement and creative task demands.

Simultaneous EEG signals from both players were recorded at a sampling rate of 500 Hz using the Neuroscan SynAmps RT 64-channel amplifier, two 64-channel Quik-Caps, and CURRY 7 acquisition software (from Compumedics, El Paso, TX). Impedances at each electrode were kept below 10 kΩ and scalp electrodes were referenced to an electrode located between Cpz and Cz.

After the EEG recording, participants filled out a demographic questionnaire capturing their musical background and improvisation experience, as well as the 40-item Empathy Quotient (EQ) (Baron-Cohen and Wheelwright 2004).

Data analysis

Behavioral data. The Max/MSP patch kept track of each block, trial, condition, notes played on each keyboard, their timing information, and success or failure of each trial. We analyzed pitch content from the performance of successful trials. For the pitch content of the performance, we extracted what notes were played during improvisation, such as number of unique melodies per block, number of out-of-key notes per phrase (notes not in the key of C major), and unique notes per phrase. For the notes played during improvisation, we define a metric for melodic similarity that expresses the pitch contour similarity between two phrases, referred to as "melodic correlation" and explained by Kim et al. (2025). We used it to express the similarity between two successive phrases from two partners (to the preceding partner's phrase, and to the following partner's phrase), the similarity of improvised phrases to the score, and the similarity between the two phrases improvised by one pianist in a trial.

For 6-note phrases, $A = \{a_1, \ldots, a_6\}$, $B = \{b_1, \ldots, b_6\}$, the melodic similarity r_{AB} is:

$$r_{AB} = \frac{1}{5}\sum_{i=1}^{5} \quad min\left(\frac{|a_i - a_{i+1}| + 0.1}{|b_i - b_{i+1}| + 0.1}, \frac{|b_i - b_{i+1}| + 0.1}{|a_i - a_{i+1}| + 0.1}\right)$$

In this calculation, we used note intervals in semitones (MIDI values are integer values expressed in semitones), and added 0.1 to avoid dividing by zero for repeated notes. This results in an unsigned value scaled between 0 and 1. If two improvised phrases are identical or one is an exact transposition of

the other, the similarity value will be 1, and if the two phrases have very different pitch contours, the similarity value will be small, but never 0. For each melody, we used notes played by the pianist (not altered auditory feedback) since we are interested in players' musical ideas, but for the other melodies played by the partner, we used the auditory feedback (including altered pitches) to account for what was heard. For melodic similarity to the previous phrase, we report results for phrase 2 and 3 only because these two phrases have a preceding phrase from the partner.

The melodic similarity and pitch content analyses included all four phrases unless noted. The melodic similarity metric was computed at each trial. In each combination of conditions (*Role* [Starter, Joiner] and *Similarity* [Same, Diff]) there were 48 trials. Thus, 48 values of the melodic similarity metric were obtained per person per condition. Pitch content metrics were computed from 96 phrases because each participant played two phrases in a trial. All metrics focusing on improvised performance were statistically analyzed in the following manner. First, we averaged values at the participant level. Second, if the data was normal (Shapiro-Wilks), then we used repeated-measures analysis of variance tests (ANOVAs) with two within-subjects factors: *Role* [Starter, Joiner] and *Similarity* [Same, Diff]. For ANOVA effect size, we report generalized eta squared, η_G^2 (Bakeman 2005). Post-hoc t-tests were run with Bonferroni corrections. If the data were not normal, then we used the Wilcox test for planned comparisons to examine the main effect of *Role* [Starter, Joiner] or *Similarity* [Same, Diff].

EEG data. The EEG data were analyzed in MATLAB (Mathworks Inc., Natick, MA) using custom scripts with functions from the Brainstorm toolbox (Tadel et al. 2011). The continuous EEG data were re-referenced using the average of all 64 electrodes. For each participant, vertical and horizontal ocular artifacts were modeled via the signal-space projection method provided in Brainstorm and removed from the continuous data. The continuous data were downsampled to 125Hz and parsed into epochs from -1500 to 4000 ms at the start of the 2nd and 3rd phrases. Channels with peak-to-peak amplitude greater than +-150 microvolts within -1000 to 3000 ms were rejected.

The time-frequency representations (TFRs) of these epochs were computed using a Morlet Wavelet (wavenumber 3) decomposition with 32 logarithmically-spaced bins from 1 to 60 Hz. Signal power in each trial was calculated by multiplying each wavelet coefficient with its complex conjugate using Brainstorm functions. The TFRs were averaged across trials separately for each participant, condition, and channel. ERD and ERS in each frequency bin were calculated as a percentage fluctuation compared to the mean power from -1000 - 3000 ms. The alpha-band ERD/ERS was then obtained by averaging frequency bins of 8-13 Hz (8.2, 9.3, 10.4, 11.7 Hz). Four electrode groups were chosen to capture alpha power in the left and right hemisphere, as well as frontocentral and parietal areas: Frontocentral (FC: F1, FZ, F2, FC1, FCZ, FC2), Centroparietal left (CPL: C1, C3, C5, CP1, CP3, CP5), Centroparietal right (CPR: C2, C4, C6, CP2, CP4, CP6), and Parieto-occipital (PO: P1, PZ, P2, PO3, POZ, PO4).

We focused our alpha power analysis on phrase 2 for the Starter and phrase 3 for the Joiner since both players are listening or playing in the previous and subsequent phrase, meaning their movements before and after the analyzed phrase are identical (Figure 1, marked with a red box). Note that this symmetrical structure for two players does not occur in phrase 1 (starting from rest) or phrase 4 (followed by unison phrase). In phrase 2, the Starter has just finished playing, which usually leads to ERS rebound after movement termination, and is now listening to the Joiner play (Pfurtscheller and Lopes da Silva 1999). At the end of this phrase, the Starter must prepare to start playing again, usually leading to ERD due to movement preparation (Pfurtscheller and Lopes da Silva 1999). In phrase 3, the same is true for the Joiner. Note that this does not apply for phrase 1 or 4, due to the movement differences before phrase 1 and after phrase 4, and the task change after phrase 4, all of which affect alpha power.

For the Starter and Joiner, we used the method described in Keil et al. (2014) to determine time windows of analysis without bias. This was completed separately for listening and playing phrases, since alpha power data are differently affected by the movements of the right hand (Pfurtscheller and Lopes da Silva 1999). For each, we averaged the four electrode groups and calculated the grand average of the difference waveform (Score - Improv) for Starter and Joiner from -1000 - 3000 ms, found the absolute

peak alpha latency within the phrase (0 - 3000 ms), and determined the time of half peak amplitude onset and offset. When alpha did not return to half amplitude by the end of the phrase at 3000 ms, the offset time was set at 3000 ms. For listening phrases, the Starter's time window was 644 - 2292 ms and the Joiner's was 988 - 3000 ms. For playing phrases, the Starter's time window was from 1951 - 3000 ms and the Joiner's was 2103 - 3000 ms. Mean alpha power was obtained using the corresponding time window separately for listening and playing, for the Starter and Joiner, across the respective time windows. We first averaged data at the participant-level and then checked for normality (Shapiro-Wilks). All alpha data passed the normality test, so we analyzed our data using repeated-measures ANOVAs with three within-subjects factors: *Task* [Score, Improv], *Similarity* [Same, Diff], and *Electrode Group* [FC, CPL, CPR, PO] and one between-subjects factor: *Pair* [1-13]. When sphericity was violated, Greenhouse-Geiser corrections were applied and corrected p-values were reported. For ANOVA effect size, we report generalized eta squared, η_G^2 (Bakeman 2005). Post-hoc tests used pairwise comparisons t-tests with Bonferroni corrections.

Regression analysis. We examined whether EQ score and improvisation experience predict improvisational behavior or alpha power. We extracted a total EQ score according to Baron-Cohen and Wheelwright (2004). For the behavioral metrics, we explored whether the pitch content of improvised melodies and melodic similarity are predicted by EQ score or years of improvisation experience. We also ran the same regression analysis with the difference between Starter and Joiner roles for each of these improvisation performance metrics.

We examined whether an individual's alpha power when both partners are assigned to improvise is predicted by EQ score or years of improvisation experience. As in the behavioral data regression, we used a total EQ score. We ran this separately during listening and playing for average alpha power over the four electrode groups. We also examined whether the difference between Starter and Joiner alpha power is predicted by EQ score or improvisation experience, to understand how individuals adapt to Starter-Joiner roles when both partners are improvising.

Statistical analysis. All statistical analyses were conducted in Python and R (R Core Team 2023). Python scripts were made for extracting MIDI numbers and keypress timing and calculating similarity between melodic phrases. R scripts with *ez* (Lawrence 2016) and *rcompanion* (Mangiafico 2023) packages were used for conducting ANOVA and pairwise comparisons. Regression analysis used the *lm* function in R.

Results

Behavioral performance of improvisation

We analyzed behavioral data of notes played and timing information to understand what musicians played during improvisation and if *Role* (Starter, Joiner), *Task* (Score, Improv) or *Similarity* (Same, Diff) had any effect.

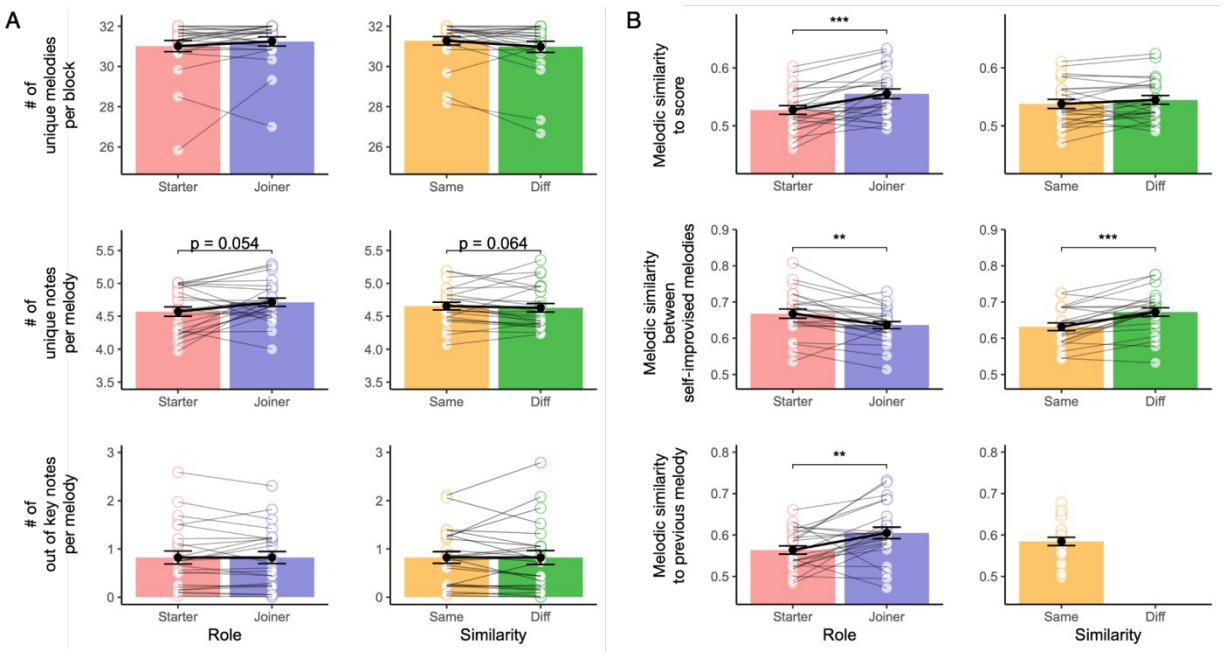

Figure 2. A. Improvisation performance metrics per participant including number of unique melodies per block (max 32), number of unique notes per melody (max 6), and number of out of key notes per melody (max 6), separated by Role and Similarity. B. Melodic similarities per participant of improvised melodies including similarity to the score, similarity between two self-improvised melodies in a trial, and similarity

to the previous melody, separated by *Role* and *Similarity*. Similarity *Same* means the partner was also improvising, whereas *Diff* means the partner played the score. Melodic similarity to the previous melody are only defined for Similarity *Same*, because otherwise the partner is playing the score. No interactions between *Role* and *Similarity* were significant. The solid black line connects mean values, where error bars express standard error of the mean (SEM), and gray lines connect circles representing each individual participant's mean. (* denotes $p < 0.05$, ** denotes $p < 0.01$ and *** denotes $p < 0.001$ by paired t-test)

Figure 2A illustrates behavioral performance in three pitch-related metrics (the number of unique notes per melody, the number of unique melodies in one bock, and the number of out of key notes per melody) through the entire session when an individual pianist served as Starter or Joiner (left), and when the duet pair's task was same or different (right). As seen in the bar plots, musicians successfully improvised such that they played a unique melody in almost every phrase of each block, and of the 6 notes in each phrase, on average no more than one note was repeated, regardless of role and task similarity. We conducted a two-way ANOVA with factors of *Role* [Starter, Joiner] and *Similarity* [Same, Diff] for these metrics. For unique notes per melody, the main effect of *Role* was marginal, $F(1, 23) = 4.13$, $p = 0.054$, $\eta_G^2 = 0.040$, as the Joiner ($M = 4.7$, $SD = 0.31$) produced more unique notes per melody than the Starter ($M = 4.57$, $SD = 0.35$) and this effect was just above the significance level. Since unique melodies per block was not normal, we used the Wilcox test for our planned comparisons, and found that *Similarity* was marginally significant ($p = 0.064$), where *Similarity* Same ($M = 31.3$, $SD = 1.0$) was larger that Diff ($M = 31.0$, $SD = 1.4$), but this difference was very small. These were the only effects in the pitch contents.

Figure 2B provides the metrics for pitch-contour similarity between two melodic phrases: one between the improvised phrase and the score, one between the two self-improvised phrases in a trial, and one to the preceding partner's phrase. We conducted the same two-way ANOVA for these similarity metrics except similarity to the previous melody, since that includes only data when both partners are improvising (*Similarity* Same). For the similarity between improvised melodies and the score, we found a

main effect for *Role*, $F(1, 23) = 18.0$, $p < 0.001$, $\eta_G^2 = 0.10$, because it was higher for the Joiner (M = 0.56, SD = 0.046) than the Starter (M = 0.53, SD = 0.039). For the similarity between the two self-improvised melodies in a trial, we found a main effect for *Role*, $F(1, 23) = 11.46$ $p = 0.0026$, $\eta_G^2 = 0.060$, where it was higher for the Starter (M = 0.67, SD = 0.071) than the Joiner (M = 0.64, SD = 0.061). This means that the Starter's improvisations were more internally consistent than the Joiner's. For the similarity between self-improvised melodies, we also found a main effect of *Similarity*, $F(1, 23) = 24.37$ $p < 0.001$, $\eta_G^2 = 0.097$, which was higher when *Similarity* was Diff (i.e. only one player is improvising; M = 0.67, SD = 0.066) than Same (i.e. both players are improvising; M = 0.63, SD = 0.064). For the melodic similarity between one's own improvised phrase and the partner's preceding improvised phrase, we found a main effect for *Role*, $F(1, 23) = 9.37$ $p = 0.0055$, $\eta_G^2 = 0.11$, where it was significantly higher for the Joiner (M = 0.61, SD = 0.068) than the Starter (M = 0.56, SD = 0.049). No interactions were significant.

EEG alpha power

Listening

We examined alpha power fluctuations as ERS/D (%) during listening (phrase 2 for the Starter and phrase 3 for the Joiner), which exhibited different shapes depending on whether a player was assigned the Score or Improv task. Time series plots and topographies are shown in Figure 3. For the Score task, alpha power increases after the start of the phrase until roughly halfway through at 1.5s, at which point it decreases. For the Improv task, alpha power at the beginning of the phrase is flat and starts decreasing around halfway through the phrase. The mound shape is most pronounced for Score conditions such that alpha increasing and decreasing slopes were steeper than Improv conditions.

In Figure 3, the mean alpha power in a time window around the alpha ERS peak, shaded in gray, is visualized in the topographies, in each duet-task condition. All the topographies consistently revealed two distinct areas of maximum ERS on the left and right, located in the bilateral centroparietal areas, which was most evident when both pianists were assigned to the score playing task (Score-Same). During score conditions, ERS also appeared in frontal areas and was smaller in occipital areas. In contrast, when

assigned to the improvisation task, topographies during listening to the partner's phrase show a weak ERS over the right centroparietal area, accompanied by ERD in the parieto-occipital midline area.

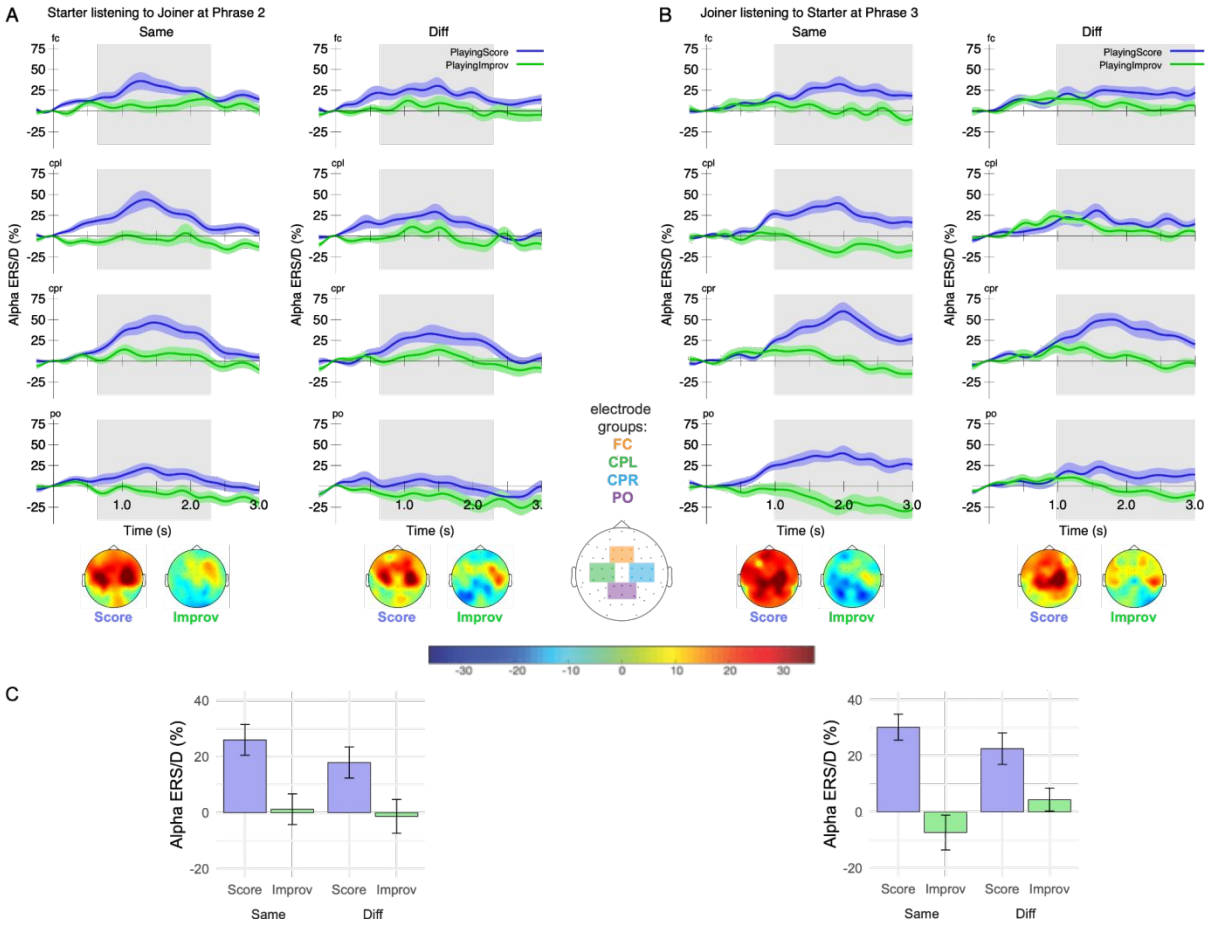

Figure 3. Alpha power time series and its topographies during listening (Starter is listening to phrase 2 and Joiner is listening to phrase 3), separated by *Task* (Score, Improv), *Similarity* (Same, Diff), and *Electrode Group* (FC, CPL, CPR, PO). Topographies for the Starter (indicated in A) correspond to the time window 644 - 2292 ms and topographies for the Joiner (indicated in B) correspond to the time window 988 - 3000 ms, which are shaded in gray on the respective time series plots. Alpha ERS/D time series for the Score is shown in blue and Improv in green. The shading around each alpha timecourse expresses SEM. C. Mean alpha power averaged across the four electrode groups by *Task* and *Similarity* for the Starter (left) and Joiner (right). Error bars express SEM.

Starter. Average alpha power from 644 - 2292 ms for the Starter was entered into separate repeated-measures ANOVAs with three within-subjects factors (*Task* [Score, Improvisation], *Similarity* [Same, Diff], and *Electrode Group* [FC, CPL, CPR, PO]) and one between-subjects factor (*Pair* [1-13]). Full ANOVA results are shown in the Supplementary Materials Table S1. There was a significant main effect of *Task*, $F(1, 13) = 16.80$, $p = 0.0013$, $\eta_G^2 = 0.18$, since mean alpha power for the Score ($M = 21.90$, $SD = 32.71$) was larger than for Improv ($M = -0.047$, $SD = 32.76$). There was also a significant main effect for *Electrode Group*, $F(3, 39) = 17.15$, $p < 0.001$, $\eta_G^2 = 0.091$, and a post-hoc pairwise t-test with Bonferroni correction confirmed that mean alpha power for CPL ($M = 11.24$, $SD = 34.07$) was significantly larger than PO ($M = -1.21$, $SD = 33.50$; $p_corrected = 0.0028$). Electrode groups CPR ($M = 18.51$, $SD = 35.52$) and FC ($M = 15.16$, $SD = 31.99$) were also significantly larger than PO, both $p_corrected < 0.001$. No significant interactions between factors were found.

Because the topographies for improvisation showed right lateralization in central areas, we compared alpha power in CPL and CPR during the Improv task assignment using a pairwise t-test, and found that mean alpha power for CPR ($M = 6.07$, $SD = 28.05$) was indeed larger than CPL ($M = -1.69$, $SD = 29.74$; $p_corrected = 0.035$). There was no significant difference between CPR and CPL for the Score task assignment.

Joiner. Average alpha power from 988 - 3000 ms for the Joiner was entered into the same ANOVA as the Starter. Full ANOVA results are shown in the Supplementary Materials Table S2. Like the Starter, there was a significant main effect of *Task*, $F(1, 13) = 29.80$, $p < 0.001$, $\eta_G^2 = 0.30$, where mean alpha power for the Score ($M = 26.38$, $SD = 31.02$) was larger than for Improv ($M = -1.46$, $SD = 31.45$). There was also a significant main effect for *Electrode Group*, $F(3, 39) = 7.65$, $p = 0.0041$, $\eta_G^2 = 0.055$, and a post-hoc pairwise t-test confirmed that mean alpha power for CPR ($M = 20.10$, $SD = 34.01$) was significantly larger than CPL ($M = 10.55$, $SD = 31.50$; $p_corrected = 0.018$) and PO ($M = 5.99$, $SD = 39.57$; $p_corrected = 0.022$). In addition, the two-way interaction *Task* × *Electrode Group* was significant, $F(3, 39) = 3.99$, $p = 0.014$, $\eta_G^2 = 0.028$, and a post-hoc t-test revealed that for the Score, mean alpha power for CPR ($M = 37.71$, $SD = 32.66$) was significantly larger than FC ($M = 22.17$, $SD = 26.38$;

p_corrected = 0.0041) and CPL (M = 22.64, SD = 28.39; *p_corrected* = 0.013), and for Improv, mean alpha power for FC (M = 4.23, SD = 30.44) was significantly larger than PO (M = -11.02, SD = 37.56; *p_corrected* = 0.030). For the three-way interaction, *Task* × *Similarity* × *Electrode Group*, $F(3, 39) = 6.84$, $p < 0.001$, $\eta_G^2 = 0.018$, a post-hoc t-test revealed that for Score, mean alpha power for Same was larger than Diff in PO (p = 0.039). This means that when a player is assigned the Score and they are listening to their partner play the score, then their alpha power is larger in occipital areas than if their partner is improvising. For Improv, mean alpha power for Same was significantly smaller than Diff in CPL ($p < 0.001$) and PO (p = 0.0084). This means that when a player is assigned the Improv task, their alpha power is lower if their partner is also improvising than if their partner is playing the score in centroparietal left and occipital areas. Overall, this indicates that Same *Similarity* enhances the alpha power difference between the Score and Improv task.

Playing

We examined alpha power fluctuations as ERS/D (%) during playing, which was phrase 3 for the Starter and phrase 2 for the Joiner. Time series plots and topographies for Starter and Joiner are shown in Figure 4. Alpha power increases after 2 seconds until the end of the phrase at 3 seconds. This increase is steeper for Improv than Score conditions, and the effect is strongest for the Joiner, where *Similarity* is Same.

The mean alpha power in a time window at the end of the phrase, shaded in gray, exhibited the topographies characterized by alpha power increase during improvisation in frontal and parieto-occipital midline areas, which is attenuated in left and right centroparietal areas. In contrast, while playing the score, frontal and parieto-occipital areas exhibited flat or decreased alpha power, which was clearest for the Joiner in the Same *Similarity*. The alpha power difference between Score and Improv conditions was clearest for the Joiner playing phrase 2 in the Same *Similarity* (Figure 4B).

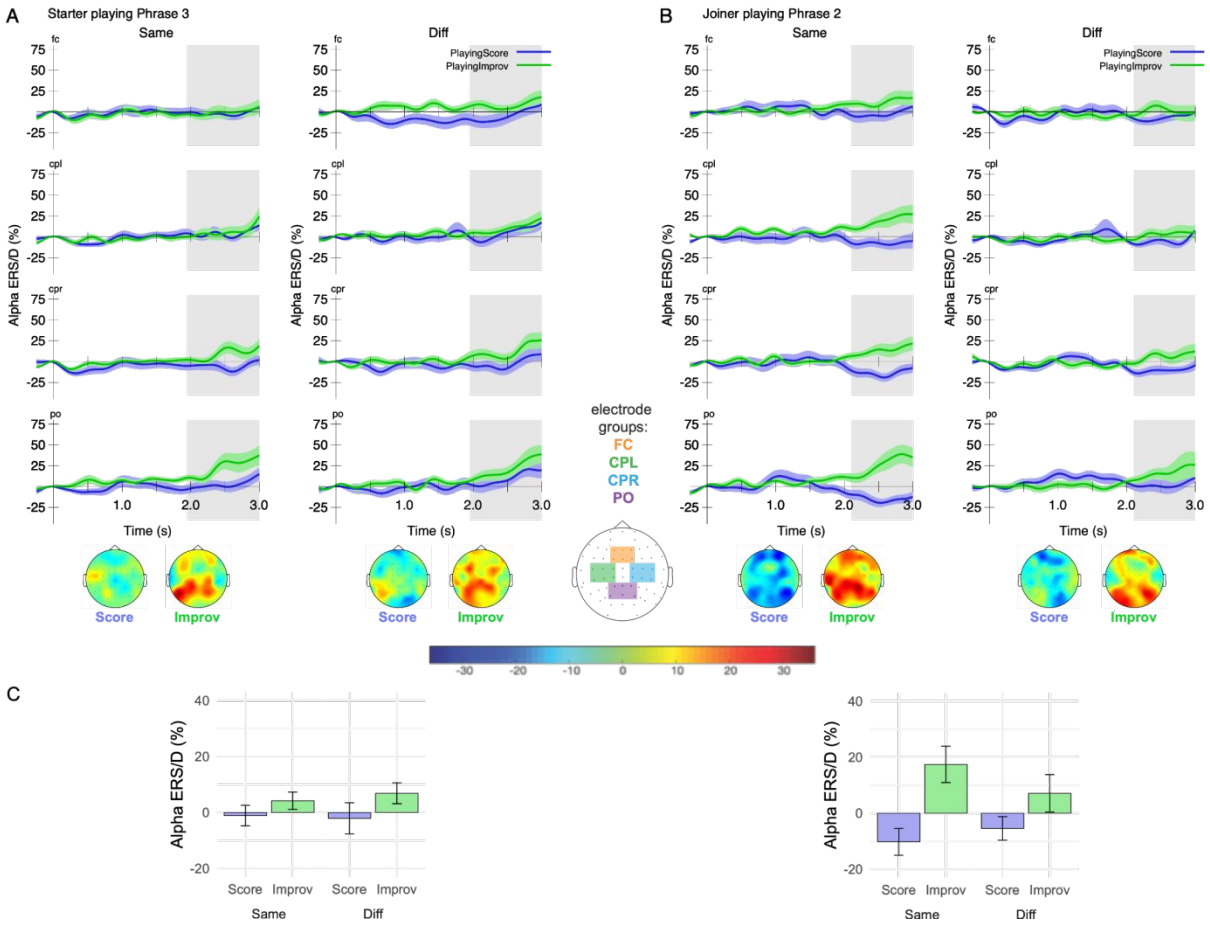

Figure 4. Alpha power time series and its topographies during playing (Starter playing phrase 3 and Joiner playing phrase 2), separated by *Task* (Score, Improv), *Similarity* (Same, Diff), and *Electrode Group* (FC, CPL, CPR, PO). Topographies for the Starter (indicated in A) correspond to the time window 1951 - 3000 ms and topographies for the Joiner (indicated in B) correspond to the time window 2103 - 3000 ms, which are shaded in gray on the respective time series plots. Alpha ERS/D time series for the Score is shown in blue and Improv in green. The shading around each alpha timecourse expresses SEM. C. Mean alpha power averaged across the four electrode groups by *Task* and *Similarity* for the Starter (left) and Joiner (right). Error bars express SEM.

Starter. Average alpha power from 1951 - 3000 ms for the Starter was entered into the same ANOVAs as listening. Full ANOVA results are shown in the Supplementary Materials Table S3. There was a significant main effect for *Electrode Group*, $F(3, 39) = 5.69$, $p = 0.0025$, $\eta_G^2 = 0.028$, and a post-

hoc pairwise t-test revealed that mean alpha power for PO (M = 7.06, SD = 26.46) was significantly larger than FC (M = -1.15, SD = 25.44; $p_corrected$ = 0.0016) and CPR (M = -0.20, SD = 24.41; $p_corrected$ = 0.026). In addition, the two-way interaction *Pair* × *Similarity* was significant, $F(12, 13) = 2.91$, $p = 0.034$, $\eta_G^2 = 0.14$, but none of the post-hoc pairwise t-tests were significant. No other main effects or significant interactions between factors were found.

Joiner. Average alpha power from 2103 - 3000 ms for the Joiner was examined via the same ANOVA as the Starter. Full ANOVA results are shown in the Supplementary Materials Table S4. There was a significant main effect of *Task*, $F(1, 13) = 9.46$, $p = 0.0089$, $\eta_G^2 = 0.16$, since mean alpha power for Improv (M = 12.23, SD = 36.94) was larger than for the Score (M = -7.86, SD = 27.26). There was also a significant main effect for *Electrode Group*, $F(3, 39) = 4.47$, $p = 0.0086$, $\eta_G^2 = 0.019$, and a post-hoc pairwise t-test revealed that mean alpha power for CPR (M = -1.21, SD = 30.09) was smaller than PO (M = 7.33, SD = 40.88; $p_corrected$ = 0.049). In addition, all three two-way interactions were significant: *Task* × *Similarity*, *Similarity* × *Electrode Group*, and *Task* × *Electrode Group*. For the two-way interaction, *Task* × *Similarity*, $F(1, 13) = 5.87$, $p = 0.031$, $\eta_G^2 = 0.027$, a post-hoc t-test revealed that for Improv, mean alpha power for Same (M = 17.40, SD = 35.86) was significantly larger than Diff (M = 7.06, SD = 37.45; p = 0.037), but there was no difference for Score. For the two-way interaction, *Similarity* × *Electrode Group*, $F(3, 39) = 3.64$, $p = 0.021$, $\eta_G^2 = 0.0096$, a post-hoc t-test revealed that for Diff *Similarity*, mean alpha power for electrode group PO (M = 9.37, SD = 40.56) was larger than FC (M = -3.04, SD = 31.71; $p_corrected$ = 0.046) and CPR (M = -1.97, SD = 28.54; $p_corrected$ = 0.050). For the two-way interaction, *Task* × *Electrode Group*, $F(3, 39) = 4.61$, $p = 0.0075$, $\eta_G^2 = 0.018$, a post-hoc t-test revealed that mean alpha power for Improv was significantly larger than Score in all electrode groups except FC; CPL: Improv (M = 10.93, SD = 34.77), Score (M = -6.83, SD = 25.76; p = 0.0047); CPR: Improv (M = 10.05, SD = 31.79), Score (M = -12.47 SD = 23.64; p = 0.0012); PO: Improv (M = 21.55, SD = 45.05), Score (M = -6.89, SD = 30.56; p = 0.0022). The three-way interaction was not significant.

Relationships between behavior, alpha, empathy score and improvisation experience

We explored whether alpha data were related to an individual's EQ score and/or improvisation experience using regression analyses (formula: Alpha ~ EQscore + ImprovYears). Alpha data of interest were both partners assigned the improvisation task for playing and listening. During listening, when both partners were assigned the improvisation task, the model was not significant. For playing phrases, the model was significant, explaining 34% of the variance as indicated by the R-squared value ($R^2 = 0.34$, $F(2, 23) = 5.85$, $p = 0.0089$) with a significant contribution of EQ score ($B = -0.68$, $p = 0.045$) and improvisation experience ($B = 1.89$, $p = 0.0086$). This means that as players' EQ score increases, their alpha power at the end of the phrase decreases, but as their improvisation experience increases, their alpha at the end of the playing phrase increases.

We further examined if these personal attributes could contribute to explaining an individual's adaptability in switching roles between the Starter and Joiner. Specifically, we calculated the alpha power within-individual difference between roles (Joiner - Starter) for both listening and playing, because when serving as a Joiner, task (Score/Improv) and task similarity (Same/Diff) affected alpha power more than when serving as a Starter.

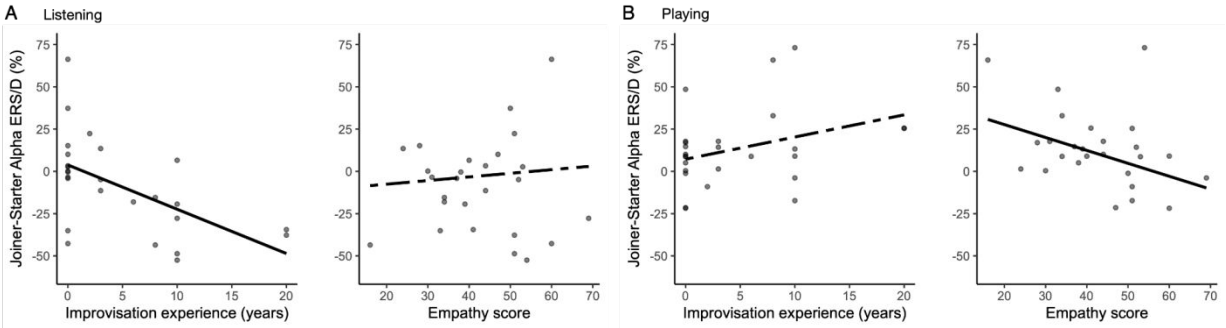

Figure 5. Regression analysis examining how within-individual alpha power difference between roles is related to improvisation experience and empathy level (formula: Alpha ~ ImprovYears + EQscore). This alpha data are taken from the blocks when both partners are improvising. A shows alpha during listening, where improvisation experience was significant in the model, but not empathy score. B shows alpha during playing, where empathy score was significant in the model and improvisation experience was marginally significant. Regression lines are solid if the variable was significant in the model and dashed if

it was not significant. Each plot shows one independent variable while holding the other fixed at the median.

As shown in Figure 5A, for listening phrases when both partners are assigned the improvisation task, the model was significant ($R^2 = 0.32$, $F(2, 23) = 5.40$, $p = 0.012$) with a significant contribution of improvisation experience ($B = 2.62$, $p = 0.0033$), but not EQ score. This means that during listening, as improvisation experience increases, the alpha power difference between Starter and Joiner roles increases, with the Joiner's alpha power lower than the Starter's. As shown in Figure 5B, for playing phrases when both partners are assigned the improvisation task, the model was significant ($R^2 = 0.26$, $F(2, 23) = 4.05$, $p = 0.031$) with a significant contribution of EQ score ($B = -0.76$, $p = 0.032$) and a marginally significant contribution of improvisation experience ($B = 1.31$, $p = 0.068$). This means that during playing, as EQ score increases, the alpha power difference between Starting and Joining roles decreases.

Since those alpha activities were associated with the EQ and improvisational experience, we also investigated whether there was any relationship between EQ, improvisation experience and the behavioral performance metrics (the number of unique melodies per block, the number of unique notes per melody, and the number of out of key notes per melody) or melodic similarity metrics (formula: Metric ~ ImprovYears + EQscore), but none of the models were significant. Similarly, we made the same regression analysis with the difference between Starter and Joiner roles for each of these metrics. None of these linear models were significant.

Discussion

We investigated alpha power timecourses during a turn-taking piano duet where pianists either played a score or improvised. Our results showed that (1) while listening to a partner's phrase, alpha resulted in a steeper increase and decrease for Score than Improv, (2) while playing one's own phrase, alpha power was flat initially, followed by an increase at the end of the phrase with a greater amount for Improv than Score, (3) for both listening and playing, alpha power modulation was larger if both players

had the same task, and for the Joining player, and (4) EQ score and improvisation experience predicted individual alpha power modulation and role-switching adaptability when improvisation is assigned to both partners.

First, we interpret that the pattern of alpha increase followed by decrease during listening is primarily related to movement (playing) before and after the listening period, in line with what has been known about the sensorimotor alpha rhythm (i.e. *mu*). That is, at the start of the listening phrase, pianists have just finished playing, and alpha ERS is consistent with rebound after termination of movement or during a rest period following ERD (Pfurtscheller 1992; Pfurtscheller et al. 1996; Babiloni et al. 2002; Klimesch et al. 2007). Then the alpha decreases around halfway through the listening phrase as pianists prepare to play again, in line with alpha ERD during movement preparation (Neuper and Pfurtscheller et al. 1996; Pfurtscheller and Lopes da Silva 1999; Klimesch 2012) and retrieval of information held in memory, especially stored motor memory traces (Krause 2006; Pesonen et al. 2006; Klimesch et al. 2007).

The assigned task of improvising before and after listening resulted in significantly flattened alpha, compared to the substantial ERS we observed when assigned to the score playing task. The reduction was observed for both Starting and Joining roles, but the Joining role (especially for joint improvisation) resulted in the greatest distinction between Improv and Score. We instructed pianists to improvise such that it would fit musically within the given context (bassline) and with each other. This requires players to listen more attentively to their partner when not playing since the partner's notes cannot be predicted (unlike the score) and they must be ready to respond in the next phrase. Thus, this improvisation-specific reduction is likely attributed to the increased attention paid to the partner's phrase and the increased encoding processing. Externally focused attention, especially in anticipation of auditory stimulus or to coordinate with a partner, has been linked to alpha ERD (Klimesch et al. 2007; Naeem et al. 2012), as well as enhanced interbrain synchrony in the gamma frequency band and mutual adaptation of timing when performing unfamiliar versus familiar music in piano duets (Gugnowska et al. 2022). Additionally, improvisers have to build an ongoing musical representation in their mind and may be

preparing what notes to play in the next phrase. Listening to the trained melody of the score activates motor and premotor cortices (Haueisen and Knösche 2001; Lahav et al. 2007), and joint music playing (scored or improvised) involves movement simulation in the partner (Novembre et al. 2014). Previous work demonstrated that motor imagery leads to alpha ERD in sensorimotor areas, which attenuates the rebound ERS (Pfurtscheller and Neuper 1997; Pfurtscheller et al. 1999; McFarland et al. 2000). These two accounts are not mutually exclusive. We suggest that the combination of the attentional demands of integrating a dynamic musical context and auditory-motor association likely explains the observed alpha reduction in the improvisation task assignment during listening.

The same idea could explain the steeper alpha increase and decrease observed in the Score task assignment. Our listening period required pianists to pay external attention to the partner's phrase while encoding and maintaining the score in working memory. With these demands, the brain would need to inhibit functions that are not the top priority at the moment. When pianists are assigned to play the score, the partner's task (Score or Improv) does not affect their own task of sticking to the score. This means less attention is needed externally. Moreover, because our score was very simple and performed repeatedly, most of the pianists could memorize it although memorization was not required. Thus, the listening period could be used for memory retention, known to cause alpha ERS (Krause 2006; Pesonen et al. 2006; Klimesch et al. 2007; Wisniewski et al. 2021). Internally focused attention may also aid retention of the score in working memory, which correlates with ERS (Pfurtscheller and Lopes da Silva 1999; Cooper et al. 2003; Cooper et al. 2006), and when the partner is improvising, would help limit distracting pitch information (Beaty 2015). Finally, the small alpha ERS in frontal areas may reflect a top-down process-specific inhibition because the partner's playing is less relevant to the listener than the player (Neuper and Pfurtscheller 2001; Cooper et al. 2003; Klimesch et al. 2007; Mazaheri et al. 2014).

During playing, alpha power remained mostly flat, followed by a small ERS at the end of the phrase. This alpha increase was steeper for Improv than Score, and the difference was only significant for the Joiner. The ERS at the end of the playing phrase may partially be related to rebound effects after movement termination, but not entirely since it begins around 2 s, slightly before the last note is played at

around 2.5 s. However, the hardest part is over by 2 s, since by then musicians have likely finished generating ideas of what notes they are going to play. Furthermore, the improvisation task of idea generation and inhibiting distractions is clearly more difficult than playing the score, so players may feel more relief after 2 s, likely explaining the steeper alpha rebound for improvisation than score playing in frontal and parieto-occipital areas, and less concentrated over sensorimotor areas. Altogether, we propose that this ERS may reflect top-down inhibition after idea generation involving the frontal cortex (Cooper et al. 2003; Klimesch et al. 2007; Fink et al. 2009a; Fink and Benedek 2014) and visual attention as well, since players would not need to look at the score or keyboards considering they are well experienced pianists. It may also be due to carryover effects from being in an internally-directed, creative state while playing that is often accompanied by ERS (Adhikari et al. 2016; Lopata et al. 2017; Fink and Benedek 2014), which may have been masked by motor movement earlier in the phrase.

At the end of the playing phrase, when both players were assigned improvisation, alpha power was positively correlated with improvisation experience, meaning that the more years of experience a player has improvising, the larger their alpha ERS will be. This is in line with previous research that found alpha power increases during improvisation only for musicians with improvisation experience (Lopata et al. 2017; Boasen et al. 2018; Sasaki et al. 2019; Rosen et al. 2020), and provides further evidence that improvisation is a distinct, trainable skill. At the same time, alpha power was negatively correlated with EQ score, which follows previous research showing greater alpha suppression for execution compared to observation correlates with larger perspective taking scores (Woodruff et al. 2011). Higher empathy scores may indicate more attention paid to the partner's reception of one's improvisation, lowering alpha power, while improvisation experience teaches players to balance attention to one's own ideas with attention to the partner, thus increasing alpha power.

For both listening and playing, most alpha power is concentrated in the left and right centro-parietal regions reflecting spontaneous central alpha (*mu*) activities, and our requirement of playing keyboards with the right hand results in less alpha power on the left hemisphere (Pfurtscheller et al. 1996; McFarland et al. 2000; Neuper and Pfurtscheller 2001; Klimesch et al. 2007). It is important to note that

in our study, the Score or Improv condition refers to one's own task, even during listening, so Improv-Diff means that the player was listening to their partner play the score, but improvising in their own preceding phrase and subsequent phrase. Thus, with our turn-taking design separating the schedules of listening and playing, we successfully identified that indeed one's own Score versus Improv task difference affected most substantially the alpha power in both listening and playing phrases. At the same time, our data clearly indicate that other factors such as participants' role (Starter, Joiner) and task similarity between partners (Same, Diff) were also influential.

Our study extends previous research by showing neural and behavioral role differences during joint musical improvisation. Duet players naturally adapted to the roles through musical turn-taking, unlike other studies which explicitly prescribed leader/follower roles. For the Joiner improvising, the alpha difference from playing the Score is larger when the partner is also improvising for both playing and listening. This is in line with prior studies showing role influences alpha (Konvalinka et al. 2014; Yoneta et al. 2022). This likely indicates that when partners exchange creative ideas, the Joiner's creative mindset is stronger, demonstrated by the significant interaction of task (score or improvisation) and similarity. Behaviorally, the Joiner is also adapting more to the Starter than vice versa, as seen in their higher melodic similarity to the preceding Starter's phrase, as well as in the lower melodic similarity between the Joiner's own two phrases in a trial than the Starter's. The results support our hypothesis and extend previous findings showing the follower role's more adaptive behavior of their tapping timing (Fairhurst et al. 2014), timing of music performance (Timmers et al. 2014; Novembre et al. 2019; Washburn et al. 2021), and body movement during music performance (Chang et al. 2017). To the best of our knowledge, we are the first to show that role differences affect melodic content of improvised musical performance.

In our improvisation task, the Joiner's musical choices in the second phrase depend largely on the musical material set by the Starter in the first phrase, necessitating more attention oriented externally toward the Starter in order to adapt. Since we provided no explicit instructions for the Starter versus Joiner roles, such adaptive improvisation behaviors and asymmetry between the roles arose naturally from

the course of the musical dialogue. Thus, 'improvisational musical choice' is sensitive to dynamically emergent musical roles in joint music making, not just explicitly pre-assigned roles. Followers' adaptive behaviors have been shown mainly in synchronization timing (Fairhurst et al. 2014; Timmers et al. 2014; Novembre et al. 2019; Washburn et al. 2021), and such timing-related behavior would be mostly unconsciously or implicitly organized. Thus, the observed alpha and behavioral patterns of musical choice here offer strong evidence of the role differences in naturalistic musical activities governed by turn-taking (e.g. playing one at a time), clearly demonstrating that call-and-response improvisational structures can engage coordinated interpersonal behavior with jointly held musical goals, without needing to play music physically simultaneously.

Lastly, we found that during listening, as players' improvisation experience increased, their alpha power difference between the Starting and Joining roles increased when both players were assigned the improvisation task. Specifically, those with more improvisation experience had lower alpha power as a Joiner than as a Starter, exaggerating the effect of alpha suppression during listening in the improvisation task for the Joining role. This could mean that experienced improvisers are more attentive to their partners in the Joining role when assigned the improvisation task, perhaps because they can more easily balance the integration of novel musical material while getting ready to improvise in the next phrase. Since previous research mainly focused on alpha power during playing improvisation (Lopata et al. 2017; Boasen et al. 2018; Sasaki et al. 2019; Rosen et al. 2020), this adds a new perspective for the listening and planning stages in a turn-taking duet improvisation, by showing that the magnitude of alpha power responses are determined not only by improvisational expertise but also performers' roles implied by the musical structure. We suggest that experienced improvisers incorporate control of their adaptive behaviors depending on their assigned musical role.

In contrast to listening, we found that during playing when both players were assigned the improvisation task, as participants' EQ score increased, their alpha power difference between Starting and Joining roles decreased. This means that higher empathy individuals had similar alpha responses as a Starter and Joiner, perhaps indicating comparable use of neural resources in both roles. In contrast,

participants with low empathy scores may be overly adapting in the Joining role, requiring more neural resources, which leads to a larger rebound at the end of the phrase due to relief from the improvisation task. This may be related to our previous piano duet studies finding individuals with higher empathy had smaller feedback-related negativity (FRN) when hearing altered pitch playback (Huberth et al. 2019; Kim et al. 2025). FRN indexes prediction error when the action outcome doesn't match the predicted outcome. The smaller FRN when encountering unexpected feedback may be caused by players inhibiting reactions in order to keep the joint performance going, which implies fewer neural resources invested during playing. In our improvisation task, the Joiner must adapt their musical ideas to fit with the Starter's previous phrase. Individuals with higher empathy may better understand the Starter's musical intention, thus investing fewer neural resources as a Joiner during playing. Admittedly, this connection between FRN and alpha related to individual empathy level is speculative. Notably, Washburn et al. (2019), which investigated alpha during a scored piano duet did not find a tangible correlation between empathy and alpha activity. Future research could investigate these connections further.

Besides our piano duet studies, we are not aware of a similar paradigm investigating alpha power, empathy, and role switching during joint improvisation. However, previous studies have found empathy and role are important for enhanced prediction and coordination of timing (Novembre et al. 2019; Washburn et al. 2019; Abalde et al. 2024) or movement (Carlson et al. 2018; D'Amario et al. 2023). One study of saxophone quartets found that higher empathy scores was related to greater alpha desynchronization during observation of their own performance, but they did not investigate role switching (Babiloni et al. 2012). Therefore, to the best of our knowledge, the current study is the first to connect individual trait empathy with the within-individual difference in alpha power responses between serving two contrasting roles.

While our study balanced design constraints within an ecologically valid musical performance, there are limitations in our design and analysis that point to future research opportunities. First, we could not directly compare alpha power in the starting versus joining roles, because we would need a longer sequence of turn-taking in each trial to even out the movement schedule differences. We chose short

phrases and trial length to limit recording time and performer fatigue based on feedback from participants in our pilot experiments. Future studies might be able to focus on how the role asymmetry changes over the course of longer turn-taking improvisation. Second, a larger sample size with substantial variability in empathy and improvisation experience might yield more informative results, considering the previously shown effects of these individual factors on interpersonal behavior (Berkowitz and Ansari 2010; Woodruff et al. 2011; Babiloni et al. 2012; Novembre et al. 2014; Lopata et al. 2017).

Additionally, it is important to note a few musical constraints in our design. First, our improvisation was limited to the pitch space, with timing restricted to eighth notes, while our paradigm tolerated slight microtiming variations. This strict timing of movements equated between score and improvisation allowed us to compare alpha power changes across participants and conditions. Still, this substantially limits how creative musicians' ideas could stretch, particularly as different harmonies and patterns of musical closure can elicit large variations in timing in naturalistic musical performance (MacRitchie et al. 2013; Gingras et al. 2016; Huberth et al. 2020; Chander et al. 2022). Lastly, although our duet study contributes to the growing body of hyperscanning studies engaging dyads to get closer to real-world musical scenarios, many improvisation contexts involve three or more performers (Wilson and MacDonald 2016), which would require a departure from a binary leader/follower distinction.

In conclusion, our investigation into alpha dynamics highlighted the important aspects of joint musical improvisation and differences from playing a score. We also found that trait empathy and improvisation experience predicted alpha power modulation in concert when both partners were assigned the improvisation task, providing evidence that not only can we capture the creative mind in real-time, but also that this mindset is cultivable. Our results also point to a new, unexplored relationship between empathy, alpha power, and individual role in the musical structure. In our data, alpha power is sensitive to externally directed attention while listening to a partner, as well as improvisational creative demands during playing. These effects are enhanced when partners' tasks are the same (both improvising or both playing score), and the joining player adapts their attention, and creative ideas to the starter, whereas the starting player is in a more stable and internal creative ideation mode throughout the trial. Importantly,

our data emphasized that joint similarity and partners' roles in the musical performance substantially influence adaptive improvisation, which occurs naturally in turn-taking. Furthermore, individual empathy was related to the alpha difference between serving starter or joiner roles. Thus, the data presented here provide novel and strong evidence that turn-taking musical conversations make players take roles accordingly and contribute to a shared goal as the musical context unfolds over time, without need of physically simultaneous improvisation or explicit role assignment. When two musicians improvise together, it requires a whole-brain process in both players, while such activities are much more subdued when one partner plays predictable material like a score. Since our alpha and behavioral data were obtained through such a simple musical improvisation task, easy enough for all participants regardless of improvisation experience, we consider that the observed results patterns successfully capture the most basic aspects of brain functions related to exercising musical creative freedom and dealing with unpredictability. Future studies could extend a similar paradigm design and examine finer details of spatio-temporal characteristics of alpha oscillatory networks underlying creative joint enterprise.

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Response Letter to the reviewers

First, we would like to thank the editor and reviewer for the feedback on our manuscript. We have addressed all of the reviewer's comments in our revision, described below in detail. As the reviewer focused mainly on the statistical analyses, the comments helped us to ensure our choice of methodologies are clearly justified in this revision. We also incorporated some of the suggested references in our Introduction wherever applicable. Updated text in the manuscript is colored in blue, like our responses here.

Reviewer: 1

I don't believe I was the primary reviewer on this manuscript, so I've approached this review with the aim of complementing the feedback already provided rather than stepping in as a full "replacement" reviewer. I've tried to keep the previous review in mind to avoid placing unnecessary burden on the authors.

We very much appreciate the reviewer's effort to be mindful about the previous review and refrain from redundancies.

This is a very engaging paper that I believe has the potential to make a substantial contribution to the literature. I do have some concerns regarding the statistical analyses, along with a few minor points that would benefit from clarification.

Thank you for recognizing the potential significance of our paper's contribution.

Major Issues:
1. A power analysis is missing.
If an a priori power analysis was not conducted, then a sensitivity analysis should be included, reporting the smallest effect size that can be detected given the current sample size (at a power of .80). Note: if the authors powered for ANOVA, then switching to mixed (as I suggest below) will not reduce their power.

Our power analysis regarding the EEG alpha power is now included in the manuscript (p. 10). We explain that our calculation is based on our previous duet-EEG alpha paper (Washburn et al. 2019) to obtain Cohen's *f*, which is further translated to ANOVA design with a power of 0.8 in G Power software.

2. ANOVA is not the most suitable tool for this question.
I agree with the previous reviewer: mixed-effects models would be a far more appropriate analysis. For example, the authors state that there were 13 pairs (26 pianists). However, the degrees of freedom in the ANOVAs indicate that the data were entered at the individual level, with "pair" treated as a between-subjects factor and dropped due to lack of significance (which can be due to the low power for the between-subject variable). This is not appropriate, as it ignores the correlation within pairs. The current analysis artificially increases statistical power by relying on the high within-pair correlation, which reduces the overall error terms for the effects

being tested. A more appropriate model would be a mixed-effects model with subjects nested within pairs, which allows the model to control for within-pair correlation.

a. Furthermore, there is little justification for averaging over trials. For more than a decade, the standard approach has been to model item (or trial) -level and pair/subject-level effects simultaneously (Baayen et al., 2008). The authors should clearly specify their model parameters, including fixed effects, random intercepts and slopes, and contrast coding, as recommended by Bates et al. (2015) and Matuschek et al. (2017). Including individual trials would also provide a small gain in power.

Thank you for providing references and feedback for the potential issue regarding ANOVA and trial averaging. Although single-trial analysis might be suitable for certain experimental designs and data sets, our main observation is neurophysiological data from EEG. Here, we follow the tradition and high standards in the EEG research field. Specifically, our analysis approach of alpha power in EEG has been conducted by averaging across trials to improve the signal-to-noise ratio, which has been the golden-standard for over 30 years (Pfurtscheller and Lopes da Silva 1999) and is still recommended today based on the renowned EEG/MEG scientists committee (Keil et al. 2014). According to this recommendation, comparisons between averages are acceptable as long as the number of trials in each average is similar (Keil et al. 2014). In our design, each subject in our study completed the same number of trials in each condition.

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Pfurtscheller G, Lopes da Silva FH. 1999. Event-related EEG/MEG synchronization and desynchronization: basic principles. Clinical Neurophysiology. 110(11):1842–1857. doi:10.1016/S1388-2457(99)00141-8.

b. Alternatively, the authors would need to follow Kenny et al. (2020), recommendations for dealing with dyadic data if they want to retain ANOVA and argue that there are no trial level issues (e.g., fatigue effects, practice effects, etc., that can be controlled by accounting for trial order in a mixed model). They would need to conduct an interclass correlation between the members of the dyad, and if the ICC value is low, they can treat the individual as independent.

Thank you for the book suggestion about dyadic data statistical analysis. However, the nature of our neurophysiological data and our research questions are very different from the topics discussed in the book regarding social science research areas, because we do not have a research design to intentionally manipulate different types of pairings as a main factor affecting our dependent variables, nor do we intend to have pairings that are task-order-specific.

Coming back to the suggestion of replacing ANOVA with the mixed-effects models, our attempt to compute them over our alpha data (4 types, starter/joiner roles x playing/listening) failed to converge, showing that the model's suitability to our dataset was limited. Therefore, in our

revision, we report the ANOVA with the update of keeping *Pair* as a between-subjects factor instead of leaving it out.

3. Post hoc t-tests duplicate the main effects from the ANOVA.
The authors often state that post hoc pairwise t-tests "confirmed" the main effects found via ANOVA. However, in a 2 × 2 within-subjects design, the ANOVA already computes an error term for each main effect and the interaction, as well as a subject-level error term (equivalent to the random intercept in mixed-effects models). Therefore, the results of a paired t-test will be mathematically identical to those of the ANOVA for the main effects. That is, $F = t^2$ in this design. An R example that shows this equivalence at the end of this review:

Thank you for pointing this out. We agree with the reviewer and have removed the redundant report of post hoc t-test results from the text.

4. Concerns about normality.
ANOVA is a special case of the general linear model (or general linear mixed model in the case of repeated measures). Like all regression analyses, it does not require the dependent variable to be normally distributed—the assumption is that the residuals are homoscedastic and approximately normal (see Cohen's Applied Multiple Regression). In the case of categorical independent variables, if the DV is normal, residuals typically are too—but this doesn't guarantee homogeneity of variance across groups. The authors' application of the Shapiro–Wilk test to such a small sample increases the likelihood of rejecting normality, prompting them to switch to nonparametric post hoc tests. This move significantly reduces power and was likely not anticipated in the original design or power estimation. This approach thus inflates Type II error without effectively protecting Type I error.

Our understanding is that checking normality before using a parametric test is still a standard procedure (Salkind 2007). We also note that typical EEG/MEG studies (solo or hyperscanning) involve a sample size of 10-30 and the data are analyzed with a similar approach (repeated measures ANOVA and post-hoc tests) in previous studies from our laboratory (Fujioka et al. 2015; Huberth et al. 2019; Washburn et al. 2019; Kim et al. 2025) and others (Lopata et al. 2017; Boasen et al. 2018; Varlet et al. 2020; Yoneta et al. 2022). We used non-parametric tests for our behavioral data comparisons whenever the Shapiro-Wilk test is significant, and we note that these tests are planned comparisons, not post-hoc.

References:
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5. Post hoc tests are underreported and overly conservative.
If only a few post hoc tests are planned, Bonferroni correction is not strictly necessary, especially with such a small sample size, and the outcome further reduces their statistical power. The authors should report the t-values, degrees of freedom, and effect sizes (Cohen's d) for each test. Reporting effect sizes is crucial for future researchers conducting power analyses. If the authors switch to mixed-effects models, then the authors do not need to think in the classical way of "post-hoc" testing and just directly code their contrasts of interest. To report effect sizes in mixed, see Westfall et al (2014).

Although Bonferroni might be criticized as conservative, our observations of main effects and interactions successfully answered our research questions by hypothesis testing. Furthermore, the Bonferroni correction for post-hoc testing was only applicable for the factors which had 3 or more levels, whereas most of our factors of interest had 2 levels (Task: score/improvisation, Similarity: same/diff). Thus, we do not think our statistical approach missed potential important findings regarding our research questions.

6. Regression models should be combined.
Regression analyses were run separately for each within-subject condition. Instead, these should be combined in mixed-effects models that include interactions between condition, empathy, and years of improvisation experience. Including all predictors in one model—run separately for each DV—would allow for better control over shared variance and lead to a more generalizable model.

We did not run regression analysis separately for each within-subject condition. Our regression analyses only examined one specific condition where both participants are assigned to improvise, in order to see if empathy quotient (EQ) score and improvisation experience have any effects on behavior or alpha power. Nevertheless, the reviewer's comment made us realize that our

description should be better explained to avoid such confusion, so we added a few clarifications in the methods section (p. 17). Also, we added figure references in the results section to help readability (p. 27).

7. Non-significant interactions still need to be reported.
In light of the replication crisis, it is no longer sufficient to simply state that an interaction was not significant. All interaction statistics should be reported, including effect sizes and confidence intervals. In a regression framework, a model summary table is sufficient; in ANOVA, this can be more cumbersome, so I suggest including this full reporting in a supplementary methods section either way.

Thank you for the suggestion. We included tables of full ANOVA results in a Supplementary Materials document and mentioned them accordingly in the results section of the text.

Minor Points:
1. Introduction (page 6, line 40):
The sentence beginning "Ideally, playing from the score..." is unclear. Why would this be the ideal way to observe power differences in improvisation? Do the authors mean "playing composed music from a score"? Given the melody's simplicity and frequency of repetition, the musicians likely memorized it, as is common in Western music performance—particularly when performers aim to focus on partner coordination.

Following the suggestion, we rephrased the sentence (p. 6).

2. Details on improvisational experience:
The methods section states that 14 subjects had improvisational experience. Was any additional information collected about the nature of this experience? For example, jazz improvisation and Baroque ornamentation rely on different strategies. Any available details would be valuable to include.

Thank you for the question. We did not collect information about the details and types of their improvisational experience. This is now mentioned in the methods section (p. 10). We agree with the reviewer that this kind of information might be more useful when systematically examining different backgrounds.

Additional literature suggestions:
To broaden the theoretical context and link to related findings, the authors might consider citing:

Improvisation & consonance:
Setzler, M., & Goldstone, R. (2020). Coordination and consonance between interacting, improvising musicians. Open Mind, 4, 88–101.
Setzler, M. D., & Goldstone, R. L. (2022). Tonal emergence: an agent-based model of tonal coordination. Cognition, 221, 104968.

Roles in musical interaction (but I have not seen much related to improv):
Demos, A. P., Carter, D. J., Wanderley, M. M., & Palmer, C. (2017). The unresponsive partner:

roles of social status, auditory feedback, and animacy in coordination of joint music performance. Frontiers in Psychology, 8, 149.
Sabharwal, S. R., Breaden, M., Volpe, G., Camurri, A., & Keller, P. E. (2024). Leadership dynamics in musical groups: Quantifying effects of musical structure on directionality of influence in concert performance videos. PLOS ONE, 19(4), e0300663. https://doi.org/10.1371/journal.pone.0300663

We thank the reviewer for suggesting these interesting articles. The studies with improvisation and musical interaction are now incorporated in the introduction.